Table 8-16. *printf-based functions*

`fprintf($stream, $format, ...$values)`	Variable number of value arguments, writes output to handle
`printf($format, ...$values)`	Variable values, sends output (e.g., terminal, or HTTP client)
`sprintf($format, ...$values)`	Variable values, returns output
`vfprintf($stream, $format, $valuesArray)`	Single array of values, writes to handle
`vprintf($format, $valuesArray)`	Single array, sends output
`vsprintf($format, $valuesArray)`	Single array, returns value

Substitution

These functions have the common mechanism of changing one byte value (or set of byte values) for another throughout the entire string.

Table 8-17. *Transform by substitution functions*

`strtolower()`, `mb_strtolower()` `strtoupper()`, `mb_strtoupper()`	Convert a limited set of ASCII printable chars to all upper or lower case
`strtr()`	Replace single-byte characters with alternative byte values, or replace byte sequences with alternatives
`substr_replace()`	Replace specified bytes with a new byte sequence
`str_replace()`	Replace all occurrences of a sequence with a new sequence
`ucfirst()`, `mb_ucfirst()` `ucwords()`, `mb_ucwords()`	Make the first (ASCII alphabet) char of a string uppercase, or the first char of all words in a string
`mb_convert_case()`	Multi-byte safe case conversion: upper, lower, sentence, etc.

`strtolower(), mb_strtolower(), strtoupper(), mb_strtoupper()`